

Uma Charitha LingaReddy

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EDUCATION

IIIT Sri City

Bachelor of Technology in Artificial Intelligence and Data Science (GPA: 8.17 / 10.0)

Sri City, India

2023 – 2027 (Ongoing)

EXPERIENCE

Canvas Space Company

AI/ML Engineer Intern

US, California

Remote

- Built multimodal pipelines integrating vision, audio, and text using transformer architectures
- Developed automated AI agents with LangGraph for workflow orchestration and multi-step execution
- Optimized model performance for efficient, scalable inference across modalities

YOKOTECH Solutions

ML Engineer Intern

Andhra Pradesh, India

Remote

- Developed a DL-based image classification system for protein crystallization (crystal, microcrystal, precipitate, clear)
- Trained and evaluated EfficientNetV2, Inception, MobileNetV2, and custom CNN using 5-fold cross-validation
- Built an end-to-end pipeline with preprocessing and feature engineering (Haralick, Gabor) with hybrid modeling

PROJECTS

Multimodal Video Understanding Pipeline | Computer Vision, Speech Processing, Generative AI

- Designed a multimodal AI pipeline for video understanding, enabling highlight detection, temporal segmentation, genre classification, and summarization using transformer-based fusion.
- Implemented fallback mechanisms using embedding similarity for robust performance on edge-case inputs.
- Tech: Hugging Face, YOLOv8, ResNet50, Whisper, Librosa, Gemini 2.5 Flash, PyTorch, Transformers.

Enhanced RIPE: PPO-based Keypoint Learning | Reinforcement Learning, Computer Vision

- Developed a PPO-based weakly supervised framework for keypoint detection and cross-view image matching without labeled data.
- Built a vision pipeline using contrastive learning and geometric verification for robust pose estimation.
- Tech: PyTorch Lightning, VGG19, Kornia, OpenCV, LO-RANSAC/MAGSAC++, PPO, Hydra

AI Placement Tutor (Multi-Agent System) | NLP, Generative AI, Agentic AI

- Built an AI-placement Tool for ATS score, resume generation, job search, cold emails, and mock interviews.
- Designed a multi-agent LangGraph + n8n workflow for automated job application and interview pipeline.
- Tech: LangGraph, n8n, Hugging Face, Streamlit, JobSpy, Docker, SQLite

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Vision Transformers (ViT), Reinforcement Learning (PPO, DQN, Actor-Critic), Model Evaluation, Cross-Validation, Mixed Precision Training

Computer Vision: Object Detection & Segmentation (YOLOv8), Image Classification (ResNet, EfficientNet), CLIP, OpenCV, Keypoint Detection, Visual Localization, Image Augmentation, Multimodal Fusion

NLP & Generative AI: Transformers (Hugging Face), LLMs (LLaMA, Gemini API), BART, Speech & Audio Processing, RAG (Retrieval-Augmented Generation), Prompt Engineering, Agentic AI

Data Science: NumPy, Pandas, Statistical Analysis, Feature Engineering, EDA, Data Visualization (Matplotlib, Seaborn, Plotly)

MLOps: Experiment Tracking, Model Versioning, CI/CD Pipelines, Model Monitoring, Hyperparameter Tuning, Pipeline Automation

Deployment & Tools: Docker, CUDA, Git/GitHub, REST APIs, Streamlit, Gradio, Hugging Face Hub

POSITIONS OF RESPONSIBILITY

Secretary: Managed mess operations, coordinated schedules, and ensured smooth dining services.

IOTA club: Contributed to cybersecurity initiatives and organized Chakravayuh during the Uthkrista Techno-Cultural Fest.